Skill Gap Analyzer using AI/ML

Submitted by: Lipsa Rani Panda
Register number: 25BCE10034

Course: Fundamentals of AI and ML
Institution: VIT Bhopal University

Date: 26.03.2026

1. Introduction

In today's competitive job market, engineering students often struggle to identify the exact skills required for different job roles. Many students prepare randomly without understanding industry expectations, which leads to inefficient learning and missed placement opportunities.

With the rapid growth of technology, there is a need for intelligent systems that can guide students in aligning their skills with job requirements. This project aims to bridge this gap using Artificial Intelligence and Machine Learning techniques.

---

2. Problem Statement

Students lack clarity about the skills required for specific job roles. As a result, they invest time in learning irrelevant technologies while missing essential skills needed for placements.

There is a need for a system that can:

- Analyze a student's current skill set
- Compare it with job requirements
- Identify missing skills
- Suggest suitable career paths

---

3. Objective

The main objectives of this project are:

- To build an AI-based system for analyzing skill gaps
- To match student skills with job roles
- To identify missing skills required for each role
- To recommend the most suitable job roles

---

4. Methodology

The project uses Natural Language Processing (NLP) and Machine Learning techniques to compare text data (skills).

Steps followed:

1. Data Collection
    - A dataset (jobs.csv) was created containing job roles and required skills
2. Data Preprocessing
    - Skills were converted to lowercase
    - Text cleaned and formatted
3. Feature Extraction
    - TF-IDF (Term Frequency-Inverse Document Frequency) was used to convert text into numerical vectors
4. Similarity Calculation
    - Cosine Similarity was used to measure similarity between student skills and job role skills
5. Output Generation
    - Match percentage calculated
    - Missing skills identified
    - Best job role recommended

---

5. Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Natural Language Processing (NLP)

---

6. Implementation

The system takes user input in the form of skills and processes it using TF-IDF vectorization. The transformed data is then compared with job role data using cosine similarity.

The output includes:

- Match percentage for each job role
- Best matching job role

- Missing skills
- Top 3 recommended roles

---

7. Results

The system successfully analyzes the skill gap between students and job roles.

Example:

Input:

python sql html

Output:

- Data Scientist: 65% match
- Web Developer: 70% match
- Missing skills identified for each role

This helps students understand what skills they need to improve.

---

8. Real-World Applications

- Career guidance systems
- Resume screening tools
- HR recruitment systems
- Skill development platforms

---

9. Challenges Faced

- Creating a realistic dataset manually
- Handling variations in skill names
- Ensuring accurate similarity calculation
- Designing a simple yet effective system

---

10. Key Decisions

- Used TF-IDF instead of complex deep learning models to keep the system simple and explainable
- Created a custom dataset to match real-world scenarios
- Focused on usability and clarity rather than complexity

---

11. Limitations

- Dataset is manually created and limited
- Does not consider experience level or projects
- Skill matching is keyword-based

---

12. Future Improvements

- Integrate real-world job datasets (LinkedIn, etc.)
- Add web interface using Streamlit
- Suggest courses for missing skills
- Use advanced NLP models for better accuracy

---

13. Conclusion

This project demonstrates how AI and Machine Learning can be applied to solve real-world problems like career guidance and skill analysis. The Skill Gap Analyzer provides a simple yet effective way to help students understand their current standing and improve their skills accordingly.

It highlights the importance of data-driven decision-making in education and career planning.

---

14. Learning Outcomes

Through this project, I learned:

- Practical application of Machine Learning concepts
- Working with NLP techniques
- Importance of data preprocessing
- Building end-to-end ML solutions
- Problem-solving and project development skills